

Exploratory Data Analysis on Global Superstore

Key Findings Report

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Dataset	Global_Superstore2.csv
Records	51,290 order lines, 24 columns
Period	January 2011 to December 2014
Total Sales	12,642,502
Total Profit	1,467,457
Overall Profit Margin	11.61%
Tools	Python, pandas, numpy, matplotlib, seaborn

This report summarizes the main results of the EDA notebook (**Global_Superstore_EDA.ipynb**). The full code, tables and all charts are inside the notebook and the images folder.

1. Data Overview and Cleaning

The dataset has 51,290 rows and 24 columns. Columns are grouped as follows:

- **Target variables:** Sales and Profit
- **Numerical:** Sales, Quantity, Discount, Profit, Shipping Cost
- **Categorical:** Ship Mode, Segment, Market, Region, Country, State, City, Category, Sub-Category, Order Priority
- **Temporal:** Order Date, Ship Date

Cleaning steps

- **Missing values:** only Postal Code had missing values (41,296 rows, 80.51%). Postal codes exist only for US orders, so the column was dropped.
- **Data types:** Order Date and Ship Date were converted to datetime. Year, Month, Quarter and Ship Days columns were added.
- **Duplicates:** 0 duplicate rows were found.
- **Outliers (IQR method):** Sales 5,655 rows (11.03%), Profit 9,755 rows (19.02%), Quantity 877 rows (1.71%). These are real large orders so they were kept. Capped columns were created for charts.

2. Statistical Summary

Column	Mean	Median	Std	Min	Max	Skew	Kurtosis
Sales	246.49	85.05	487.57	0.44	22,638	8.14	176.73
Quantity	3.48	3	2.28	1	14	1.36	2.28
Discount	0.14	0	0.21	0	0.85	1.39	0.72
Profit	28.61	9.24	174.34	-6,600	8,400	4.16	291.41
Shipping Cost	26.38	7.79	57.30	0	933.57	5.86	50.02

- Sales and Shipping Cost are strongly right skewed. A few very large orders pull the mean far above the median.
- Profit has very high kurtosis, meaning heavy tails on both sides (big profits and big losses).
- Correlations: Sales and Shipping Cost 0.77, Sales and Profit 0.48, Discount and Profit -0.32.

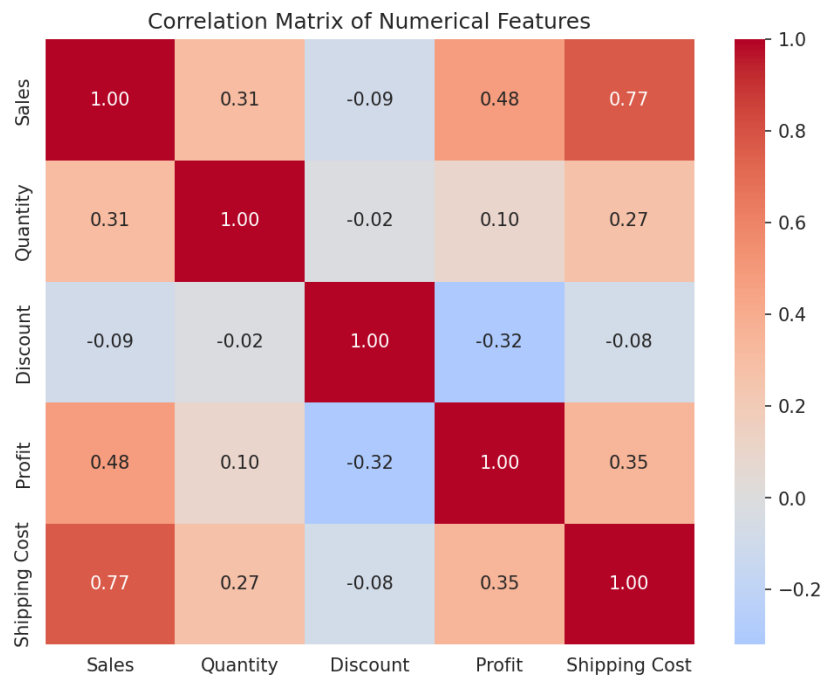


Figure 1: Correlation matrix of numerical features

3. Visualizations

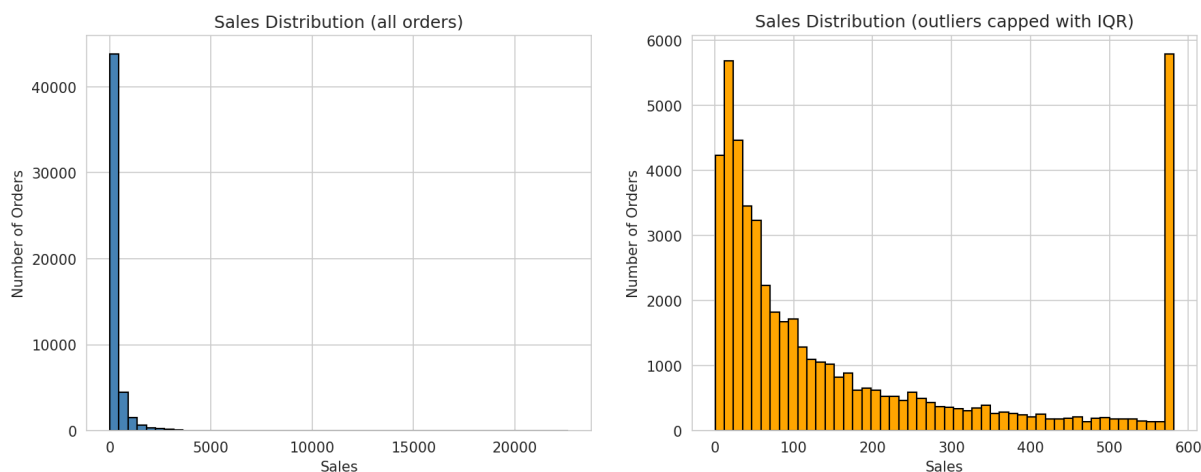


Figure 2: Sales distribution (raw and IQR-capped)

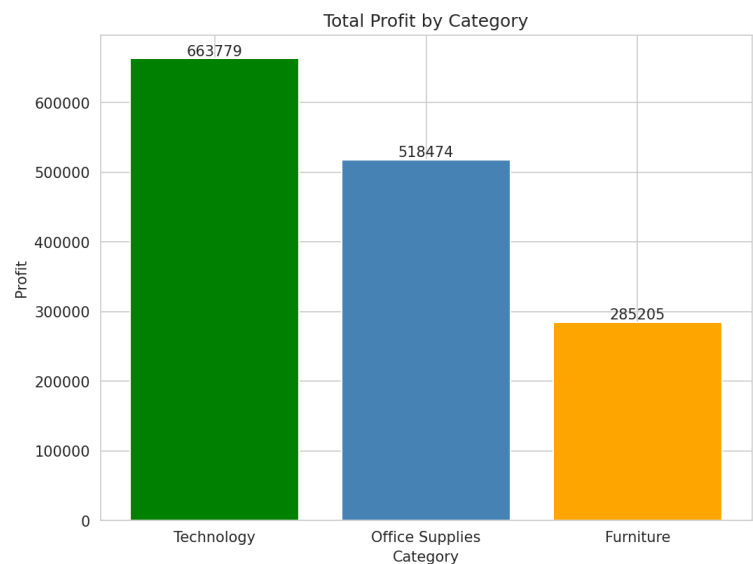


Figure 3: Total profit by category

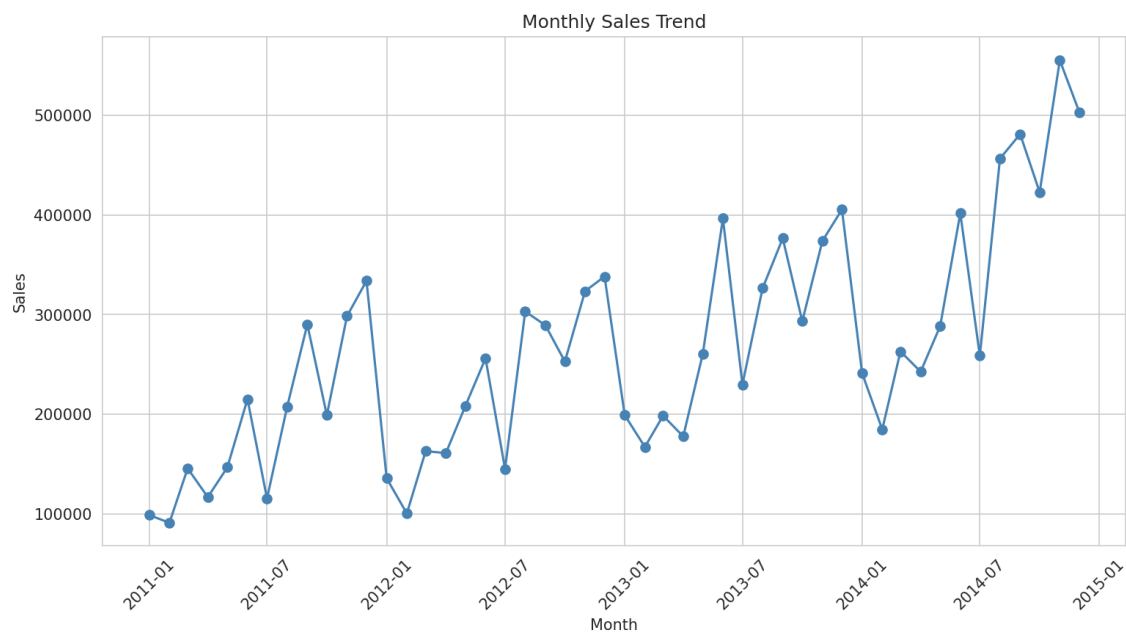


Figure 4: Monthly sales trend

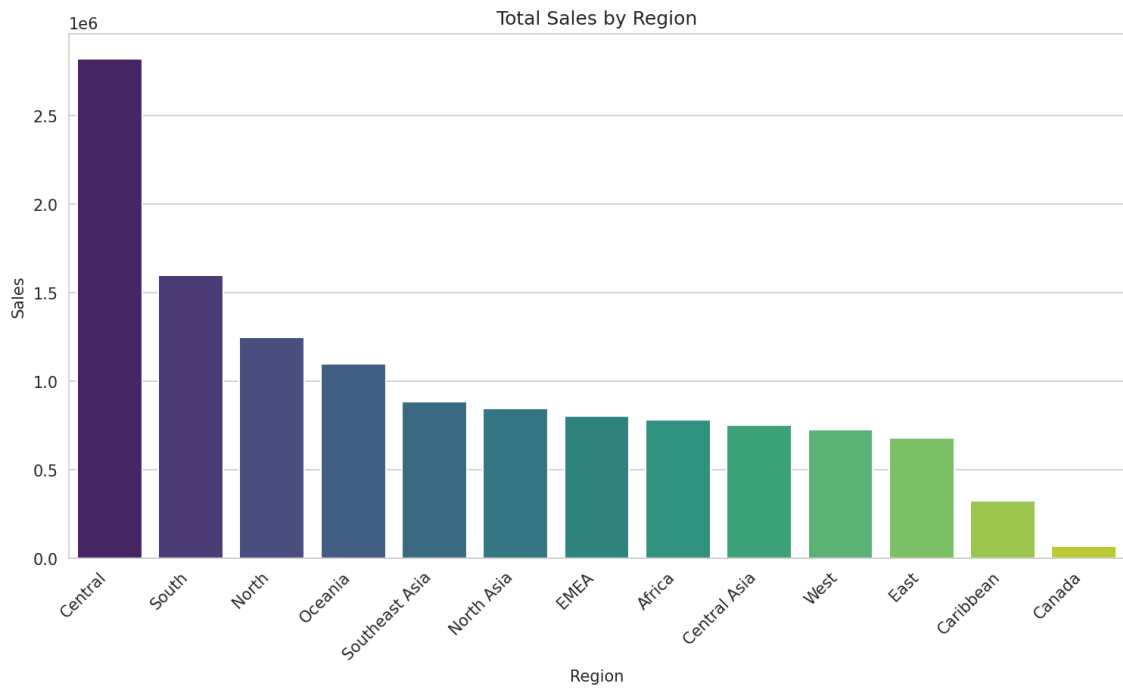


Figure 5: Total sales by region

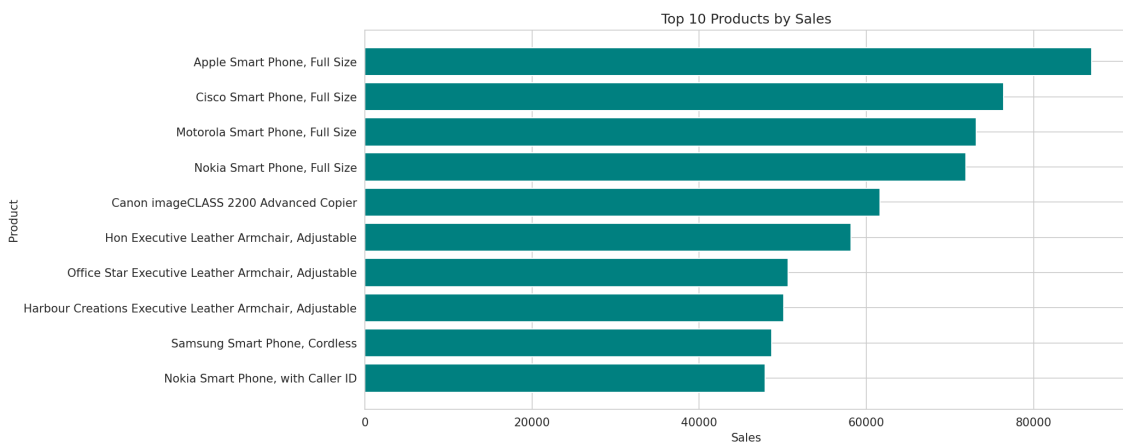


Figure 6: Top 10 products by sales



Figure 7: Discount vs profit



Figure 8: Quantity by shipping mode

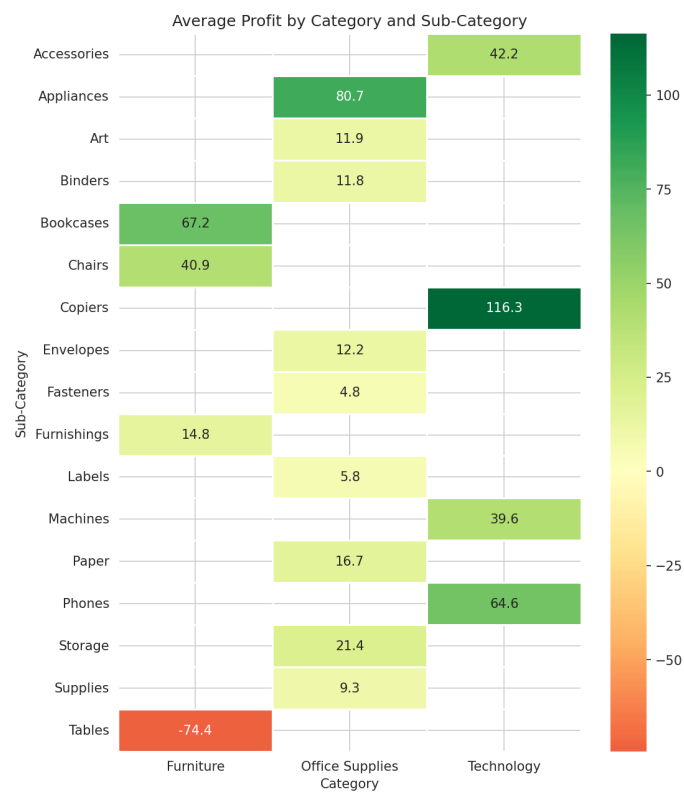


Figure 9: Average profit by category and sub-category

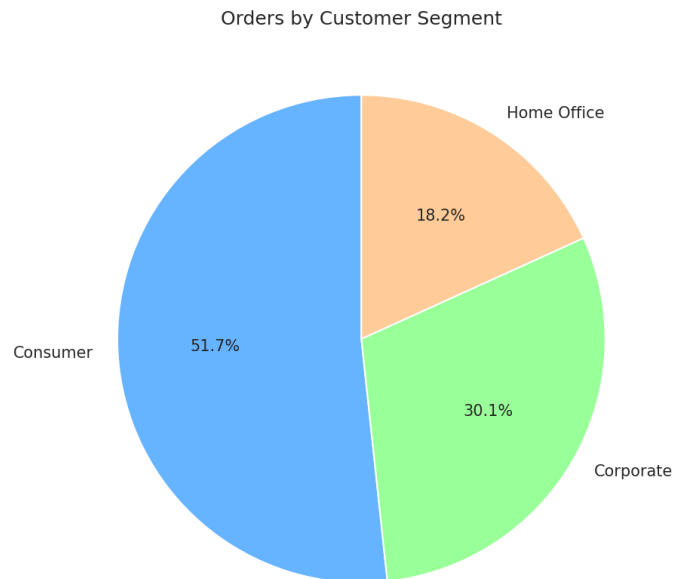


Figure 10: Orders by customer segment (Consumer 51.7%, Corporate 30.1%, Home Office 18.2%)

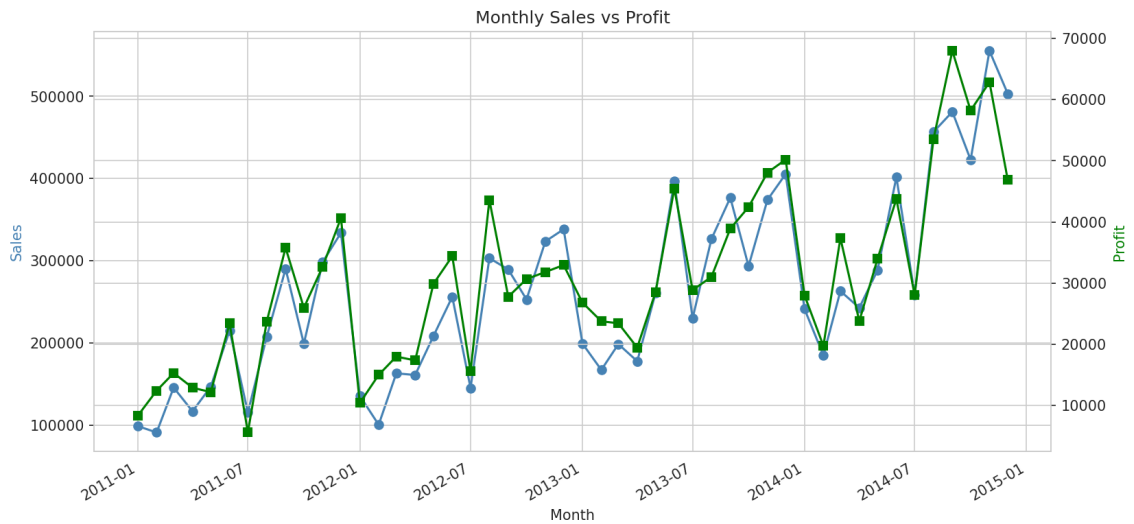


Figure 11: Monthly sales vs profit (dual axis)



Figure 12: Total profit by sub-category

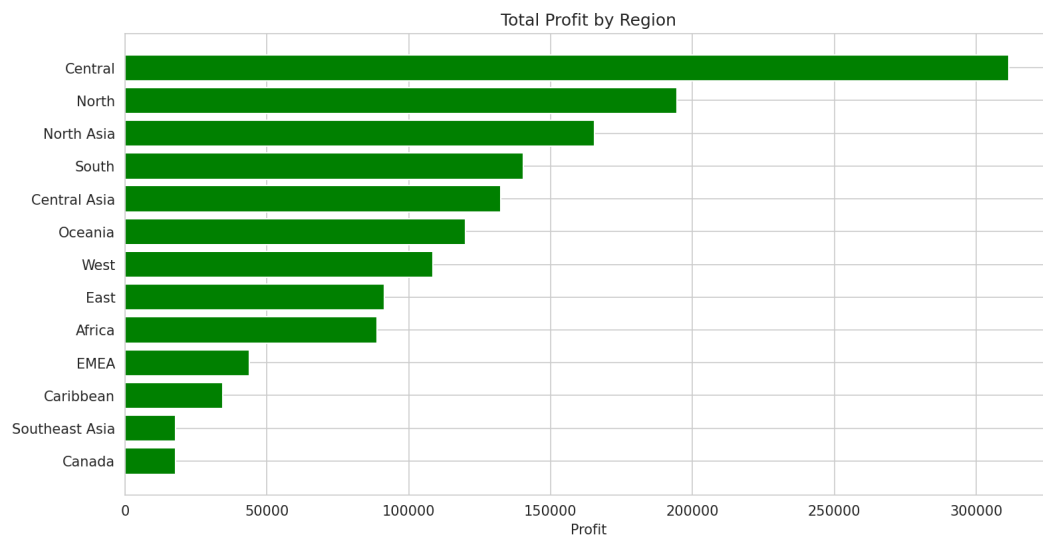


Figure 13: Total profit by region

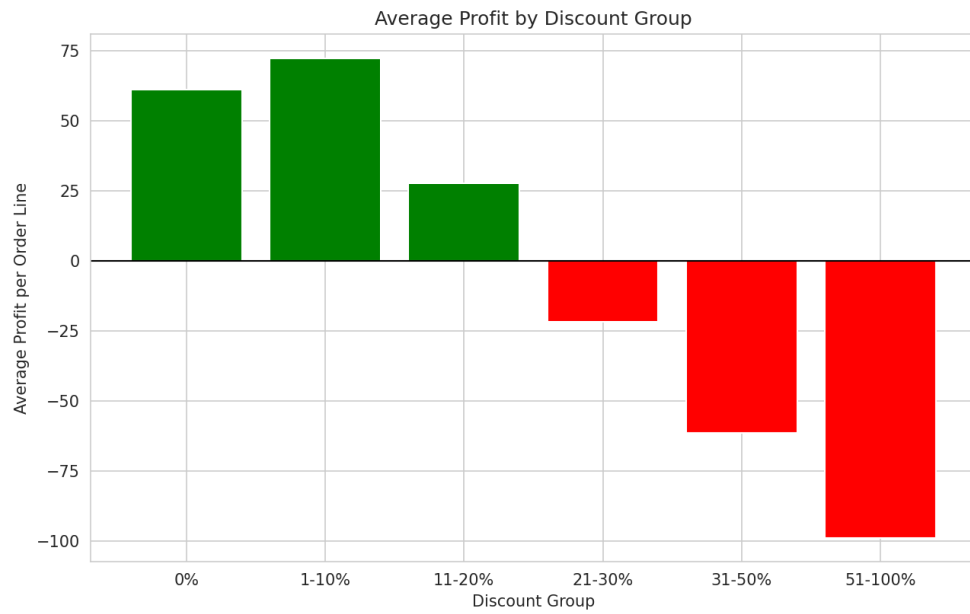


Figure 14: Average profit by discount group

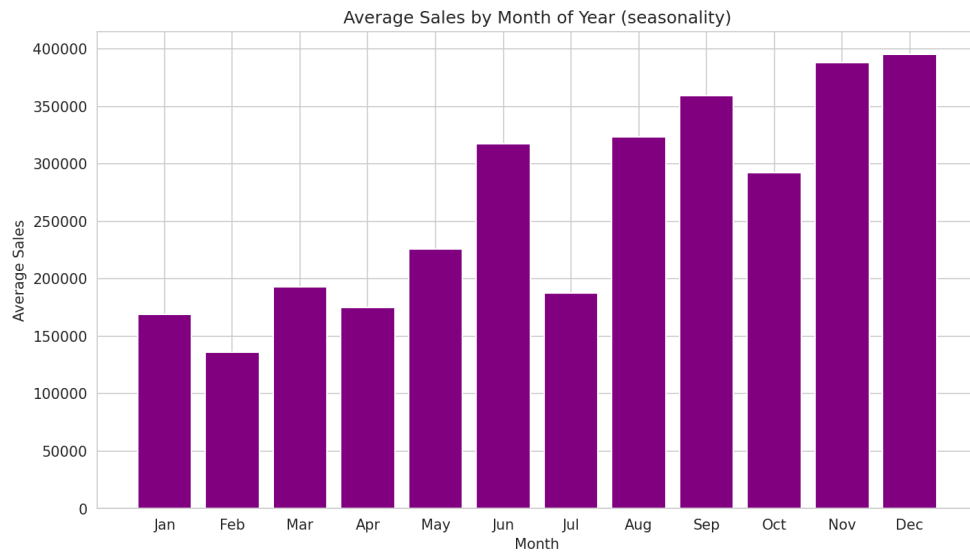


Figure 15: Average sales by month of year

4. Key Insights

Top 3 most profitable categories

Category	Sales	Profit	Margin %
Technology	4,744,558	663,779	13.99
Office Supplies	3,787,070	518,474	13.69
Furniture	4,110,874	285,205	6.94

Top sub-categories: Copiers (258,568), Phones (216,717) and Bookcases (161,924). **Tables** is the only loss making sub-category (-64,083).

Underperforming regions

Region	Sales	Profit	Margin %
Southeast Asia	884,423	17,852	2.02
EMEA	806,161	43,898	5.45
South	1,600,907	140,356	8.77

No region loses money in total, but these three earn much less than the overall 11.6% margin. At country level, Turkey (-98,447), Nigeria (-80,751) and the Netherlands (-41,070) lose the most.

Discount and profit

Discount	Avg Profit	Total Profit	Loss rows %
0%	61.04	1,770,695	0.0
1-10%	72.28	338,189	19.3
11-20%	27.61	173,255	23.3
21-30%	-21.88	-21,156	62.2
31-50%	-61.55	-380,945	87.4
51-100%	-98.89	-412,582	100.0

Correlation between Discount and Profit is -0.32. Profit turns negative above 20% discount, and every order line above 50% discount loses money.

Seasonal pattern

- Q4 gives 34.0% of yearly sales, Q3 27.5%, Q2 22.7% and Q1 only 15.7%.
- Highest months: December, November and September. Lowest: February and January.
- Sales grew from 2.26M (2011) to 4.30M (2014). Profit grew from 249K to 504K.

5. Business Recommendations

- **Cap discounts at 20%.** Discounts above 30% cost about 794K in profit. Add an approval rule for anything higher.
- **Fix or drop the Tables sub-category.** It loses 64K with an average discount of about 29% and pulls down the Furniture margin (6.9%).

- **Review Southeast Asia, EMEA and loss making countries** (Turkey, Nigeria, Netherlands). Check shipping cost, discounts and product mix before investing more.
- **Focus on Technology and Q4.** Technology earns the most profit and Q4 brings a third of sales, so stock and campaigns should be ready by September. Use small discounts in slow months.
- **Reduce shipping cost.** Same Day (about 43) and First Class (about 41) cost double Standard Class (about 20). Encourage Standard shipping on bigger orders and charge extra for fast delivery.

Conclusion

The business grows every year and is profitable overall. The main problems are heavy discounting, the Tables sub-category and a few weak regions. Fixing these would improve profit without needing extra sales.

Limits: data covers only 2011 to 2014, and profit is measured per order line.